

Spring 2008, Math706 – Numerical Linear Algebra

Homework 2

1. Let $\mathbf{A}$ be an $m \times m$ matrix, let a_j be its j -th column. Prove the *Hadamard's inequality*:

$$|\det \mathbf{A}| \leq \prod_{j=1}^m \|a_j\|_2.$$

2. Let $\mathbf{A}$ be a matrix with the property that columns $1, 3, 5, 7, \dots$ are orthogonal to $2, 4, 6, 8, \dots$. In a reduced QR factorization $\mathbf{A} = \hat{\mathbf{Q}}\hat{\mathbf{R}}$, what special structure does $\hat{\mathbf{R}}$ possess?
3. Determine the eigenvalues, determinant, and singular values of a Householder reflector. Additionally, for the eigenvalues, also give a geometric interpretation.
4. Consider the 2×2 orthogonal matrices

$$F = \begin{bmatrix} -c & s \\ s & c \end{bmatrix}, \quad J = \begin{bmatrix} c & s \\ -s & c \end{bmatrix},$$

where $s = \sin \theta$ and $c = \cos \theta$ for some θ . The first matrix F is a Householder reflector, and the second one J effects a rotation and is called *Givens rotation*.

- (a) Describe exactly what geometric effects left-multiplications by F and J have on the plane $\mathbb{R}^2$.
 - (b) Describe an algorithm for QR factorization that is analogous to Algorithm 10.1 in Page 73, but based on Givens rotations instead of Household reflection.
 - (c) Count the number of floating points involved the new algorithm proposed in (b) and compare it with that of Algorithm 10.1.
5. Suppose the $m \times n$ matrix $\mathbf{A}$ has the form

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_1 \\ \mathbf{A}_2 \end{bmatrix},$$

where $\mathbf{A}_1$ is a nonsingular matrix of dimension $n \times n$ and $\mathbf{A}_2$ is an arbitrary matrix of dimension $(m - n) \times n$. Prove that $\|\mathbf{A}^+\|_2 \leq \|\mathbf{A}_1^{-1}\|_2$.

6. (a) Write a MATLAB function $[\mathbf{Q}, \mathbf{R}] = \text{mgs}(\mathbf{A})$ that computes a reduced QR factorization $\mathbf{A} = \hat{\mathbf{Q}}\hat{\mathbf{R}}$ of an $m \times n$ matrix $\mathbf{A}$ with $m \geq n$ using the Gram-Schmidt orthogonalization.
- (b) Write a MATLAB function $[\mathbf{W}, \mathbf{R}] = \text{house}(\mathbf{A})$ that computes an implicit representation of a full QR factorization $\mathbf{A} = \mathbf{Q}\mathbf{R}$ of an $m \times n$ matrix $\mathbf{A}$ with $m \geq n$ using Householder reflectors. The output variable $\mathbf{W}$ is a $m \times n$ matrix whose columns are the vectors v_k defining the successive Householder reflections. Then

write a MATLAB function $\mathbf{Q} = \text{form}(\mathbf{W})$ that takes the matrix $\mathbf{W}$ produced by **house** as input and generates a corresponding orthogonal matrix $\mathbf{Q}$.

Print and test your programs by the reduced QR factorization of $\mathbf{Z}$ where

$$\mathbf{Z} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 7 \\ 4 & 2 & 3 \\ 4 & 2 & 2 \end{bmatrix}.$$

Also compare your results with the MATLAB's built-in command $[\mathbf{Q}, \mathbf{R}] = \text{qr}(\mathbf{Z}, 0)$ and comment on any difference you see.

7. Let us consider the polynomial least squares fitting problem. Define t to be the m -vector corresponding to linearly spaced grid points from 0 to 1. Define $\mathbf{A}$ to be the $m \times n$ matrix associated with least squares fitting on t by a polynomial of degree $n - 1$, and $\mathbf{b}$ to be the function $\cos(4t)$ evaluated on this grid. Set $m = 50$ and $n = 12$. Calculate and print (to sixteen-digit precision) the least squares coefficient vector $\mathbf{x}$ by the following methods:

- (a) QR factorization computed by **mgs**.
- (b) QR factorization computed by **house**.
- (c) QR factorization computed by **qr**.
- (d) Solution of the normal equation $\mathbf{A}^* \mathbf{A} = \mathbf{A}^* \mathbf{b}$ using MATLAB's command `\`.

Comment on any differences you observe.